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Lowering Transportation Emissions for a Sustainable Future

Abstract

Climate change is increasingly affecting communities through rising temperatures, drought, wildfire risk, and deteriorating air quality. Although these challenges are global, local transportation choices contribute directly to greenhouse gas emissions and air pollution. In California, transportation is a major source of greenhouse gas emissions, making changes in how people travel an important component of climate action. This paper examines how Torrance, California, can reduce transportation-related emissions through expanded public transportation, improved walking and bicycling infrastructure, and increased adoption of electric vehicles. These strategies can reduce greenhouse gas emissions while also improving air quality, mobility, and community health. Because transportation systems influence everyday behavior, local governments have an important opportunity to make sustainable choices more accessible. By investing in cleaner transportation and infrastructure, Torrance can contribute to California's broader climate goals while creating a healthier and more sustainable community.

Keywords: climate change, transportation emissions, Torrance, electric vehicles, public transportation, sustainable transportation

Introduction

Climate change is no longer a distant environmental concern. Communities are already experiencing its effects through increasing temperatures, drought, wildfire risk, and changes in air quality. California is particularly vulnerable to these impacts, making the reduction of greenhouse gas emissions an important environmental and public-health priority. While climate change requires action at the national and international levels, local communities also play an important role because many emissions-producing activities occur as part of everyday life.

Transportation is particularly important because people rely on cars and other vehicles for commuting, school, shopping, and other daily activities. Gasoline-powered vehicles release carbon dioxide and other pollutants into the atmosphere. According to the U.S. Environmental Protection Agency (EPA), a typical passenger vehicle produces approximately 4.6 metric tons of carbon dioxide annually, although emissions vary depending on fuel economy and miles driven. Gasoline vehicles also produce pollutants such as nitrogen oxides and particulate matter that contribute to poor air quality.

For a city such as Torrance, reducing transportation emissions represents a practical opportunity to address climate change at the local level. Torrance can reduce transportation-related emissions by strengthening public transportation, expanding safe infrastructure for walking and bicycling, and supporting the transition to electric vehicles. These approaches can complement one another by reducing

the number of vehicle trips, providing alternatives to gasoline-powered transportation, and making cleaner transportation more convenient for residents.

The Environmental Impact of Transportation

Transportation contributes to climate change primarily through the combustion of fossil fuels. When gasoline and diesel are burned in internal-combustion engines, vehicles release carbon dioxide, a major greenhouse gas that contributes to atmospheric warming. The EPA reports that carbon dioxide accounts for approximately 99 percent of greenhouse gas emissions from vehicle tailpipes. Gasoline-powered vehicles also produce pollutants including nitrogen oxides, carbon monoxide, particulate matter, and other smog-forming compounds.

The environmental impact of transportation extends beyond what comes directly from a vehicle's tailpipe. Producing and distributing gasoline requires extracting petroleum, transporting it, refining it, and delivering the finished fuel to consumers. Each stage can generate additional greenhouse gas emissions. Consequently, reducing gasoline consumption can address both direct vehicle emissions and emissions associated with fuel production.

Individual transportation decisions therefore have cumulative consequences. A single short car trip may produce a relatively small amount of pollution, but millions of vehicle trips collectively create substantial emissions. This makes transportation an especially important sector for climate policy because changes in travel behavior can produce benefits across an entire community.

Expanding Public Transportation

One of the most effective strategies for reducing transportation emissions is making public transportation more convenient and accessible. Buses and other forms of mass transit can transport multiple passengers during the same trip, potentially reducing the number of individual vehicles on the road. Torrance already has an established public transportation system, creating a foundation on which the city can build.

The City of Torrance's Climate Action Plan specifically identifies increased transit use as a strategy for reducing greenhouse gas emissions from mobile sources. The plan recommends increasing transit service, frequency, and speed and identifies bus rapid transit as one potential approach. According to the city, improving transit can reduce the number of vehicles needed to accommodate transportation demand and consequently lower carbon dioxide emissions.

Torrance could build on these strategies by improving the frequency and reliability of bus service, increasing connections between major destinations, and making routes easier for residents to understand and access. Cleaner buses, including electric buses, could further reduce local air pollution. Encouraging residents to substitute public transportation for some automobile trips would not require eliminating cars; rather, it would provide residents with a practical alternative when driving is unnecessary.

Improving Walking and Bicycling Infrastructure

Reducing transportation emissions does not depend solely on changing vehicles. It also requires changing the infrastructure that determines how easily people can choose different forms of transportation. Walking and bicycling produce no direct transportation emissions and can be particularly effective alternatives for short-distance trips.

However, people are less likely to walk or bicycle when sidewalks are incomplete, intersections are difficult to cross, or bicycle routes do not feel safe. Torrance could encourage active transportation by expanding protected bicycle lanes, improving pedestrian crossings, maintaining sidewalks, and providing shade along heavily traveled walking routes. Such improvements could make walking and bicycling more practical for students, families, and residents traveling short distances.

A community bike-sharing system could provide another option for short trips. Residents who do not own bicycles would be able to use them when needed without purchasing or storing a personal bicycle. Although walking and bicycling alone cannot replace every automobile trip, increasing their use can reduce unnecessary vehicle travel while providing additional benefits such as physical activity and more pedestrian-oriented public spaces.

Supporting Electric Vehicle Adoption

Another important strategy is transitioning from gasoline-powered vehicles to electric vehicles (EVs). Unlike conventional gasoline vehicles, battery-electric vehicles produce no tailpipe emissions. Although electricity generation can produce emissions, the EPA reports that total emissions associated with driving an electric vehicle are typically lower than those of gasoline vehicles, particularly when electricity is generated from renewable sources.

Electric vehicles therefore represent an important component of a broader transportation-emissions strategy. However, simply replacing every gasoline vehicle with an electric vehicle is not the only solution. EVs still require energy and materials, and they do not address traffic congestion or the amount of land devoted to automobiles. A more comprehensive approach combines vehicle electrification with public transportation, walking, and bicycling.

Torrance can support EV adoption by expanding access to charging infrastructure, particularly in locations where residents may not have convenient access to private charging. Public charging stations can reduce a major barrier to EV ownership by making charging more accessible. The city could also consider policies that encourage cleaner vehicles while ensuring that transportation improvements remain accessible to residents across different income levels.

A Comprehensive Local Strategy

The most effective approach to transportation emissions is not a single intervention but a combination of strategies. Public transportation can provide an alternative to driving, walking and bicycling can replace short automobile trips, and electric vehicles can reduce emissions from trips that still require a car.

This approach is consistent with Torrance's existing climate planning. The city's Climate Action Plan established a goal of reducing community greenhouse gas emissions by 49 percent from 2005 levels by 2035. Achieving ambitious reductions will require continued attention to transportation because mobility is closely connected to residents' daily routines.

Importantly, transportation policies should focus on accessibility rather than simply asking individuals to change their behavior. If public transportation is infrequent, bicycle infrastructure is unsafe, or charging stations are difficult to access, residents may have limited practical alternatives to driving. Local government can therefore influence emissions by making sustainable choices easier, safer, and more convenient.

Conclusion

Climate change is a global challenge, but meaningful solutions can begin within individual communities. Transportation is an especially important area for local action because daily travel decisions contribute to both greenhouse gas emissions and air pollution. In Torrance, reducing transportation emissions does not require eliminating cars. Instead, the city can create a transportation system in which residents have more opportunities to choose cleaner alternatives.

Expanding reliable public transportation, developing safe walking and bicycling infrastructure, and supporting electric vehicle adoption can work together to reduce emissions and improve the quality of life for residents. Torrance has already identified transportation and transit as important components of its climate strategy, providing a foundation for further action.

Sustainability depends on making environmentally responsible choices possible on a large scale. A bus ride, bicycle trip, or electric vehicle may appear to be a small change, but when thousands of residents have access to these alternatives, the cumulative effect can be significant. By rethinking how people move through the city, Torrance can reduce pollution, support healthier communities, and contribute to the broader effort to create a sustainable future.

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